

The subproduct-system dimension question was answered negatively

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Status: literature already answered (full negative answer). Gerhold–Skeide’s Corollaries 5.5–5.6 give an explicit obstruction and counterexample to the question of Shalit–Solel.

1 Exact source question

Shalit and Solel define subproduct systems and, in Section 6.2 of arXiv:0901.1422v3, ask whether every function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying

$$f(m+n) \leq f(m)f(n) \quad (m, n \in \mathbb{N}) \tag{1}$$

is realized by a subproduct system X with $\dim X(n) = f(n)$ for all n . The question is on printed/PDF page 35:

SUBPRODUCT SYSTEMS

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Here is an interesting question that we do not know the answer to: *given a solution $f : \mathbb{N} \rightarrow \mathbb{N}$ to the functional inequality*

$$f(m+n) \leq f(m)f(n) \text{ , } m, n \in \mathbb{N},$$

does there exists a subproduct system X such that $\dim X(n) = f(n)$ for all $n \in \mathbb{N}$?

Remark 6.8. One can cook up simple examples of subproduct systems that are not standard. We will not write these examples down, as we already know that such a subproduct system is isomorphic to a standard one.

6.3 Representations of subproduct systems. Fix a W^* -semigroup E .

The inequality (1) is plainly necessary because a subproduct system has an isometric coproduct map $X(m+n) \hookrightarrow X(m) \otimes X(n)$. The question is whether it is sufficient.

2 The later answer

Gerhold and Skeide prove in Theorem 3.15 of arXiv:1402.0198v2 that a sequence $(d_n)_{n \geq 0}$ is the dimension sequence of a finite-dimensional discrete subproduct system if and only if it is the

cardinality sequence of a word system, also called a factorial language. Corollary 5.5 then gives the strictly stronger necessary family

$$d_{m+n+k} \leq d_{m+k}d_{n+k} \quad (m, n, k \in \mathbb{N}). \quad (2)$$

In particular,

$$d_{k+1} \leq d_k^2 \quad (k > 0). \quad (3)$$

Their immediately following Corollary 5.6 explicitly concludes that not every submultiplicative sequence is realizable. Both statements and their counterexample appear on printed/PDF page 14:

$$\begin{aligned} &= ((a_1, \dots, a_{m+k}), (a_{m+1}, \dots, a_{m+n+k}), (a_{m+n+1}, \dots, a_{m+n+\ell+k})) \\ &= (\text{id}_{Y_m} \times j_{n,\ell})((a_1, \dots, a_{m+k}), (a_{m+1}, \dots, a_{m+n+\ell+k})) \\ &= ((\text{id}_{Y_m} \times j_{n,\ell}) \circ j_{m,n+\ell})(a_1, \dots, a_{m+n+\ell+k}) \end{aligned}$$

proves associativity for $m, n, \ell \geq 1$. For the cases involving $m = 0$ or $n = 0$ or $\ell = 0$ there is nothing to prove. So the $Y_n = X_{n+k}$ together with the maps $j_{m,n}$ form a Cartesian system. $\square$

5.4 Corollary. Suppose for $n \in \mathbb{N}$ there is a function $f: \mathbb{N}_0^{n-1} \rightarrow \mathbb{N}_0$ such that for every word system $X^\triangleright$ we have

$$\#X_n \leq f(\#X_1, \#X_2, \dots, \#X_{n-1}).$$

Then for every Cartesian system $X^\triangleright$ we have

$$\#X_{na+b} \leq f(\#X_{a+b}, \#X_{2a+b}, \dots, \#X_{(n-1)a+b}).$$

Proof. By the preceding two theorems the $Y_n = X_{na+b}$ form a Cartesian system. $\square$

5.5 Corollary. For every dimension sequence $d_n = \#X_n$ of a word system $X^\triangleright$, we have

$$(5.3) \quad d_{m+n+k} \leq d_{m+k}d_{n+k}$$

for all $m, n, k \in \mathbb{N}$. In particular, $d_{k+1} \leq d_k^2$ for every $k > 0$.

Proof. Equation (5.3) follows, because the sequence d_n is submultiplicative ($d_{m+n} \leq d_m d_n$). The formula $d_{k+1} \leq d_k^2$, in the case $k = 1$ is directly submultiplicativity, and in the case $k > 1$ follows from (5.3) with $m = n = 1$. $\square$

5.6 Corollary. Not every submultiplicative sequence d_n is the cardinality sequence of a word system.

Proof. A cardinality sequence fulfills $d_3 \leq d_2^2$. However, the sequence $d_1 = 2, d_2 = 1, d_3 = 2$, and $d_k = 0$ for $k > 3$ is submultiplicative, but $d_3 \not\leq d_2^2$. $\square$

Of course, from Corollary 3.16 it follows that all corollaries remain true for Cartesian systems, and from Theorem 3.15 it follows that all corollaries remain true for subproduct systems replacing cardinality sequence with dimension sequence.

Thus the answer to the source question is *no*.

3 Proof intuition and exact verification

For a factorial language, let W_j be its set of admitted words of length j , so $d_j = |W_j|$. Every admitted word of length $m + n + k$ determines two overlapping admitted factors: its first $m + k$ letters and its last $n + k$ letters. The two factors overlap in exactly k letters, so together they recover the original word. Hence

$$W_{m+n+k} \longrightarrow W_{m+k} \times W_{n+k}$$

is injective, proving (2). The word-system/subproduct-system dimension equivalence transfers this inequality to all subproduct systems.

The counterexample in Corollary 5.6 is

$$d_1 = 2, \quad d_2 = 1, \quad d_3 = 2, \quad d_j = 0 \quad (j > 3).$$

It is submultiplicative: the only positive target terms beyond d_1 are d_2 and d_3 , and

$$d_2 = 1 \leq d_1^2 = 4, \quad d_3 = 2 \leq d_1 d_2 = d_2 d_1 = 2;$$

every later target is zero. But (3) with $k = 2$ would require $2 = d_3 \leq d_2^2 = 1$, a contradiction.

If $\mathbb{N}$ in the source question is interpreted as excluding zero in the codomain, use instead

$$d_0 = 1, \quad d_1 = 2, \quad d_2 = 1, \quad d_3 = 2, \quad d_j = 1 \quad (j \geq 4).$$

This sequence is positive and submultiplicative. Indeed, sums 2 and 3 give exactly the two checks above, while every term at an index at least four equals 1 and every product on the right is at least 1. It still violates $d_3 \leq d_2^2$. The negative answer is therefore independent of the convention for natural numbers.

4 Provenance, scope, and review

This is categorized as *literature already answered*, not as a new counterexample. The later paper cites Shalit–Solel, studies exactly the possible fiber-dimension sequences, proves their equivalence with factorial language complexity sequences, and explicitly states the negative submultiplicativity corollary. The positive-tail modification above only removes a harmless notation convention.

The result settles the yes/no realization question. It does not classify all realizable sequences; Gerhold–Skeide reduce that broader classification to the generally open combinatorics of factorial languages.

The bounded status check covered the run indexes, the local parsed corpus, exact question/phrase searches, and arXiv searches for subproduct systems, dimension sequences, and submultiplicative sequences. Human review need only confirm the index convention and the transfer supplied by Theorem 3.15 of the supporting paper.

References

- [1] O. M. Shalit and B. Solel, *Subproduct systems*, Documenta Mathematica **14** (2009), 801–868; arXiv:0901.1422v3.
- [2] M. Gerhold and M. Skeide, *Subproduct systems and Cartesian systems; new results on factorial languages and their relations with other areas*, Journal of Stochastic Analysis **1** (2020), no. 4, Article 5; arXiv:1402.0198v2; doi:10.31390/josa.1.4.05.